INNOVATION CHALLENGE 2026 · THEME: AI FOR GOOD

PRAHARI प्रहरी

AI Digital Arrest Shield

Sub-theme 5 — **AI for Smart Forensics & Public Safety**

REAL-TIME COERCION DETECTION | ON-DEVICE | INTERVENTION, NOT WARNING

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THE PROBLEM

“Digital arrest” does not exist anywhere in Indian law.

Criminals impersonating police, CBI or RBI hold a victim on a video call for hours, forbid them from hanging up or telling anyone, and take their savings by UPI — where the money **cannot be recalled**.

₹644 cr

LOST IN 2025

₹1,918 cr at the 2024 peak

17,264

REPORTED CASES

1,23,672 in 2024

46%

RUN FROM ABROAD

Cambodia · Myanmar · Laos

Cases fell 86% in 2025 — and that is the argument for this product.

The fall was not produced by better detection. It was produced by a national **awareness** campaign — SMS blasts, caller tunes, Mann Ki Baat. What worked was interrupting victims and telling them what was happening to them. Prahari does that per call, at the moment of risk, instead of hoping they remember a broadcast. ₹644 crore a year is still taken from the people it did not reach.

Source: Indian Cyber Crime Coordination Centre (I4C) / Ministry of Home Affairs, reported February 2026. Wider context — 28.15 lakh cybercrime complaints and ₹22,495 crore lost across all categories in 2025.

WHY DETECTION IS NOT THE PROBLEM

Every phone can already warn you. Almost nothing interrupts you.

What exists today

Google, Truecaller and every major bank already flag suspicious calls. Scam call detection is a solved, commoditised problem — and people still lose the money.

Why it fails anyway

The scam's entire purpose is to make the victim distrust their own judgement. By the time money moves, a man in a police uniform has spent ninety minutes telling them their family will be arrested tonight.

A notification is not a match for ninety minutes of coercion. Any product that stops at "⚠️ this may be a scam" has already lost.

So we did not build a warning. We built an interruption.

THE TECHNICAL CLAIM

It does not match keywords. It counts co-occurring attack stages.

Every scam sentence is something a real person could say — a genuine officer really does say *"this is Inspector Sharma from Cyber Crime"*. What no genuine officer ever does is say that **and** forbid you from hanging up **and** ask you to move money.

1	**Authority impersonation**	police · CBI · RBI · TRAI
2	**Fear induction**	FIR · laundering · non-bailable
3	**Isolation**	"do not tell your family"
4	**Control & surveillance**	AnyDesk · fake courtroom
5	**Money extraction**	"refundable" verification

STAGES PRESENT	MULTIPLIER
1 — a conversation	1.00×
3 — a pattern	1.40×
5 — a crime	1.90×

Trust signals subtract

Genuine callers offer a callback number, state they will never ask for an OTP, and tell you to take your time. Each of those **lowers** the score. This is why it stays silent on real bank calls.

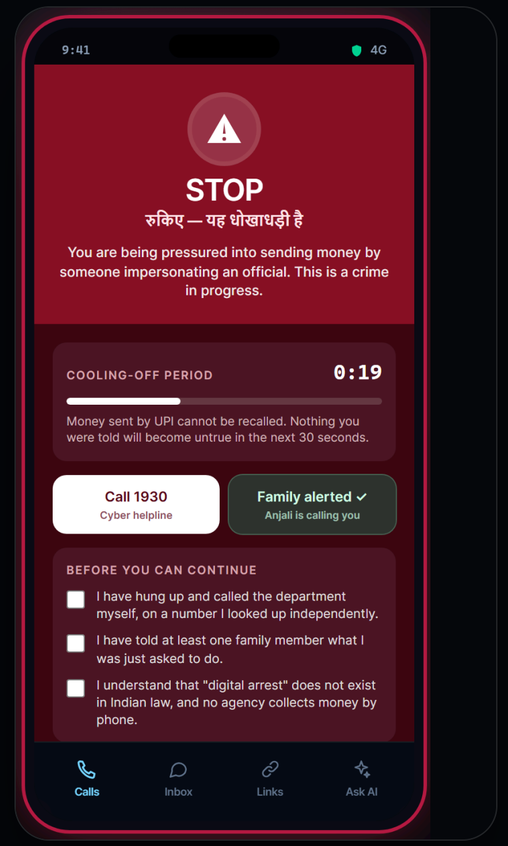
THE CORE INNOVATION

The Pause Protocol.
It takes the screen. It does not ask.

- **A 30-second enforced cooling-off.** The dismiss button is disabled. Not a suggestion — a lock.
- **One tap to 1930**, the real cybercrime helpline, and one tap to a family member with live context about what is happening.
- **A three-point verification checklist** that must be completed before the screen can be dismissed at all.
- **Elderly Protection Mode alerts the guardian automatically** — no action required from a victim who is, by design, too frightened to act.

Why thirty seconds works

Nothing the caller said becomes untrue in thirty seconds. But the scam runs on **speed and isolation** — and this removes both. The guardian gets a ringing phone, not a message. A scammer can talk over a message.



MEASURED, NOT ASSERTED

136 labelled cases. Zero false alarms.

100%
PRECISION
0 of 61 genuine cases flagged

80%
RECALL
60 of 75 frauds caught

25/25 · 12/12
LINKS · EMAILS
separate corpus, 100% both

The corpus is adversarial on purpose

61 genuine cases include 24 deliberate hard negatives — a real KYC reminder, a real customs notice, a real FIR acknowledgement, and fraud-awareness messages that quote the scam script **verbatim**. Talking about fraud is not fraud.

And it is not an independent benchmark

We wrote both the detector and the corpus, so we treat it as an instrument, not a score. Its value was what it exposed: the first run scored **24% recall** and revealed four whole scam families the engine had no patterns for. The 15 remaining misses are oblique phrasings — the case for the semantic layer on the roadmap.

`npm run eval` · `npm run eval:surfaces` · `npm run calibrate` — reproducible from the repo

PRIVACY, ETHICS, AND WHAT WE DID NOT BUILD

The detection engine never touches the network.

- **No audio is transmitted or written to storage.** Only a rolling text transcript in memory, discarded when the call ends.
- **Gemini is deliberately not load-bearing.** It explains decisions the on-device engine has already made. Pull the network cable and Prahari still detects, still interrupts, still alerts the guardian.
- **Never asks for an OTP, PIN, password or Aadhaar number** — anywhere in the product. It tells users to stop and verify independently.
- **Recording affects both parties**, so production requires an explicit consent screen and a persistent recording indicator.

Deliberately not built: voice deepfake detection

A classifier we could not train and validate properly in 36 hours would produce **false accusations against real family members** — worse than not shipping it. It is on the roadmap, not in the demo.

Also stated honestly

Bank transaction interception is shown as the Pause Protocol screen, not a live integration — that needs a partner. The adversarial simulation sends audio to Google; the protection path does not.

BUSINESS MODEL & ROADMAP

The buyer is not the victim. It is their daughter in another city.

CHANNEL	WHY IT WORKS
Family safety plan	The adult child has the payment method and the motivation
Bank & fintech licensing	Banks carry the reimbursement liability
Telecom bundling	Network-level value-added service
Freemium consumer	Free detection; Guardian Circle on subscription

Next, in order

1 · Android dev build — speakerphone mic capture, `NotificationListenerService` for SMS and WhatsApp, Accessibility Service on UPI payment screens.

2 · On-device speech — Whisper-small, Vosk for Hindi. Restores the privacy guarantee end to end.

3 · Semantic layer for oblique phrasings, and continuous ingestion of I4C advisories.

4 · Voice deepfake detection, once it can be validated.

> **Detection is commoditised. Intervention is not.**
>
> Prahari · प्रहरी — built for the ten minutes before the transfer.